PENGZHEN LI（李鹏振）

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EDUCATION

University of Tennessee, Knoxville expected 2027
Ph.D. in Economics
Committee: James Lake (Chair), Georg Schaur, Matthew Van Essen, Timothy Fitzgerald

Shandong University, Jinan 2018
M.A. in Finance — thesis published in *Technological Forecasting and Social Change* (2019); 1,000+ citations

Tongji University, Shanghai 2015
B.A. in Economics

RESEARCH INTERESTS

International Trade, Environmental and Energy Economics, Applied Econometrics, and Political Economy.

PEER-REVIEWED PUBLICATIONS

Four SSCI/SCI-indexed publications; 1,132 total citations, h-index 4 (Google Scholar, August 2026).

"Do green technology innovations contribute to carbon dioxide emission reduction? Empirical evidence from patent data." K. Du, **P. Li**, and Z. Yan. *Technological Forecasting and Social Change*, 146 (2019), 297–303. **1,000+ citations (Google Scholar).**

"Spatial optimization of the sustainable aviation fuel supply chains from forest residues via fast pyrolysis/hydrotreatment considering feedstock ash content variability." **P. Li**, T. E. Yu, N. Labbé, N. Abdoulmoumine, M. Garcia-Perez, K. P. Hoyt, and B. C. English. *Biomass and Bioenergy*, 208 (2026), 108793.

"Assessing the impact of preprocessing and conversion technologies on the sustainable aviation fuel supply from forest residues in the Southeast USA." **P. Li**, T. E. Yu, C. Trejo-Pech, J. A. Larson, B. C. English, and D. N. Lanning. *Transportation Research Record*, 2678(9) (2024), 639–654.

"Beef price spread relationship with processing capacity utilization." C. Martinez, **P. Li**, C. N. Boyer, T. E. Yu, and J. G. Maples. *Journal of the Agricultural and Applied Economics Association*, 2(1) (2023), 133–145.

WORKING PAPERS

Measured Quality Falls Within the Firm: Market Access and Chinese Exports [Job Market Paper] — evidence from Chinese customs records, 2000–2014. [paper]
Abstract: This paper links the 2002 removal of tariff uncertainty on Chinese exports to the United States to a decline in measured quality within continuing export relationships. Comparing a firm's US and non-US shipments of the same product, measured quality falls by about 6 percent among cells that ship in both periods in the three years before an overlapping quota removal, and by about 11 percent in the full sample over the whole window, which contains that removal. Restricting the sample to cells that ship in both periods retains 83 percent of the full-window estimate and yields a decline twice as large as the full-sample estimate in the three clean years. The price and quantity

responses are not those of a markup cut on an unchanged good, and the decline is not the input-tariff upgrading of the same years, which varies only within firm and year and is absorbed by the design. We develop a model in which improved access can extend a seller's reach to buyers below a sorting boundary, who lower the return to appeal. Under stated conditions, appeal is capped and can fall. The continuing-cell price and quantity pattern is consistent with this mechanism but does not identify it against other sources of a demand-residual shift.

Trade Restrictiveness Indices with Discriminatory Tariffs with James Lake. *Presented at the Midwest Economic Theory and International Trade Meetings, Virginia Tech, 2025.* [paper]
Abstract:

Estimating Mean-Field Games with a Payoff Threshold: Candidates and Party Lists in Brazil with Matt Van Essen and Luiz Lima. *2024 J. Fred and Wilma Holly Award (best second-year paper).* [paper]
Abstract:

Price Risk, Irreversible Investment, and the Welfare Benchmark for the Section 45Z Clean Fuel Production Credit with T. Edward Yu. [paper] [slides]

How Should 45Z Pay for Risk? A Welfare-CVaR and Project-Finance Decomposition of the IRA Clean Fuel Production Credit with T. Edward Yu. [paper] [slides]

Estimating Pork Primal Price Relationships Using a Threshold Vector Autoregressive Analysis with Charles Martinez, Christopher N. Boyer, T. Edward Yu, and David Anderson. *SSRN working paper 4121953 (2022).* [paper]

WORK IN PROGRESS

A Weight-Free Benchmark for Multilateral Tariff Bargaining with Georg Schaur. *Draft, August 2026.* [paper] [slides]

PRESENTATIONS

Midwest Economic Theory and International Trade Meetings, Virginia Tech — Spring 2025
University of Tennessee, Economics Brown Bag Seminar Series — Spring 2024, Fall 2024
University of Tennessee, Applied Micro Group — Spring 2024, Fall 2024
Western Regional Science Association, Annual Meeting, Hawaii — 2023
Western Agricultural Economics Association, Annual Meeting, Santa Fe, NM — 2022
Southern Agricultural Economics Association, Annual Meeting, New Orleans, LA — 2022

PROFESSIONAL EXPERIENCE

University of Tennessee – Knoxville, TN — 2021 – present
Graduate Research & Teaching Assistant. USDA NIFA and FAA ASCENT projects (2021–2023): mixed-integer supply-chain optimization (GAMS), IMPLAN/DEA siting analysis, and spatial data pipelines (ArcGIS, R, Python) for forest-residue sustainable aviation fuel; two publications and three working papers.

Shandong University – Jinan, China — 2016 – 2018
Graduate Research Assistant. Panel-threshold and machine-learning analysis of green innovation and CO_2 emissions (published in *TFSC*, 2019).

TEACHING EXPERIENCE

University of Tennessee

Instructor

ECON 351: Monetary Economics	Spring 2025, Spring 2026
ECON 322: The Global Economy: Trade and Development	Summer 2025
ECON 313: Intermediate Macroeconomics	Summer 2026

Teaching Assistant

Ph.D. courses:	
ECON 511: Microeconomic Theory	Fall 2023, Fall 2024, Fall 2025
ECON 512: Advanced Microeconomics	Spring 2024
ECON 581: Mathematical Methods in Economics	Summer 2024
Master's courses:	
AREC 525: Agribusiness Operations Research Methods	Spring 2022, Spring 2023

HONORS AND AWARDS

Charles B. Garrison Award for Excellence in Teaching (best teaching), University of Tennessee	2026
Summer Graduate Assistantship, University of Tennessee	2025
J. Fred and Wilma Holly Award (best second-year paper), University of Tennessee	2024
J. Fred and Wilma Holly Fellowship, University of Tennessee	2023 – present
Graduate Assistantship, University of Tennessee	2021 – present
Academic Scholarship for Master's Students, Shandong University	2017
Freshman Scholarship, Shandong University	2016
Outstanding Rainbow Volunteer, Tongji University (tutoring pupils in Shanghai)	2013 – 2015
National Endeavor Scholarship, Tongji University	2014
Tongji Scholarship of Excellence, Third Prize, Tongji University	2014

SKILLS

Programming Languages: Stata, R, Python, SQL, GAMS, MATLAB, LaTeX, ArcGIS, IMPLAN.
Methods: panel and causal econometrics (FE, IV, DiD, event study, SCM, GMM, panel threshold), time series (TVAR, VECM, mGARCH), mixed-integer and stochastic programming, DEA, cooperative and mean-field games, machine learning.
Credentials: CFA Level II.
Languages: Chinese (native), English (fluent).

REFERENCES

James Lake (Chair)
Associate Professor of Economics
Department of Economics
University of Tennessee
Email: jlake7@utk.edu

Georg Schaur
Leigh Burch III Economics Professor and Haslam Family Faculty Fellow
Department of Economics
University of Tennessee
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T. Edward Yu
Professor of Agricultural and Resource Economics
Department of Agricultural and Resource Economics
University of Tennessee
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